

THE STORY OF STEAM POWER

HOW WATER VAPOR WAS UTILIZED TO DRIVE THE INDUSTRIAL REVOLUTION

Although the power of steam was not harnessed until the 17th century, scientists had understood its potential for hundreds of years. As far back as the 1st century CE, the Greek scientist Hero of Alexandria had discussed a device—the aeolipile—that illustrated the possibilities of water vapor.

The aeolipile worked by heating water in a mounted sphere that had two bent nozzles. When steam was released through the nozzles, the sphere would rotate. Although it had no practical use at the time, this was the first indication of experiments with steam power. More dramatic developments took place in the 17th century, when the first boiler was invented. Although it was little more than a pressure cooker, from this point onward, a steady stream of innovations followed.

POWERING INDUSTRY

By the 18th century, engineers had realized how steam-powered devices could be used to pump water out of mines—an important

issue in light of the growing demand for coal in Europe during the Industrial Revolution. Scientists soon realized that steam could also be used to power engines. Thomas Newcomen had invented a steam engine in 1712, but it was the improvements made by James Watt that made the device more efficient. Watt's key innovation consisted of condensing steam, so that the engine did not need to heat and cool the cylinder, making it far more efficient. Soon, steam power was being used to fuel ships and locomotives, enabling them to travel farther and faster. By the 19th century, it was being used to produce electricity, something that continues to the present day, using much of the technology developed over the preceding centuries.



Richard Trevithick
In addition to developing the world's first steam railroad locomotive, the English engineer Richard Trevithick also adapted his high-pressure engine for use in iron mills and steam-powered barges.

“IN THE WHOLE HISTORY OF **TECHNOLOGY** IT WOULD BE DIFFICULT TO FIND **A GREATER SINGLE ADVANCE** THAN THIS.”

L. T. C. Rolt, English writer and engineer, *Thomas Newcomen: The Prehistory of Steam*, 1963

1819

Crossing the Atlantic
The US vessel *Savannah* becomes the first ship to cross the Atlantic using steam power as well as sails. The era of sails ends soon after.



Savannah steamship

1867

The water-tube boiler
In the US, George Babcock and Stephen Wilcox invent the water-tube boiler, in which water circulates in tubes. It is used to make electricity in 1882.



Babcock and Wilcox steam boiler

Early 20th century
Geothermal power
Scientists in Lardarello, Italy, discover “geothermal,” or “dry steam,” energy and build the first power station of its kind in 1911.



Geothermal power station

1829

Stephenson's “Rocket”
English engineer George Stephenson applies steam power to locomotives, and his “Rocket” becomes a commercial success.



Stephenson's “Rocket”

1884–97

The steam turbine
Sir Charles Algernon Parsons develops a steam turbine generator, which produces huge amounts of electricity. It is used to power large ships, such as the *Titanic*.



The *Titanic* powered by Parson's steam turbine

20th century

Steam turbines and nuclear power
Controlled nuclear chain reactions create heat in reactors, which boils water to produce steam and drive a steam turbine in order to produce electricity.